

Plasmonic Nanoantennas for Helicity-Dependent All-Optical Switching

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Next generation magnetic recording devices are likely to incorporate optical methods for magnetisation reversal at femtosecond speeds and nanometre scales using circularly polarised light. Such processes require a helicity-resolved plasmonic element that can focus circularly polarised laser light to a sub-wavelength position. Using a simulation approach, this study seeks to explore suitable designs for a plasmonic antenna capable of achieving all-optical switching at the standard bit size currently expected from hard drive devices.

Index Terms—Ferrimagnets, Helicity, Multiphysics, Plasmonics.

I. INTRODUCTION

ALL -optical control of ferrimagnetic multilayers opens a path toward next-generation magnetic storage. The discovery by Stanciu et al. [1] of deterministic magnetisation switching in synthetic GdFeCo using circularly polarised light sparked interest in ultrafast magnetic dynamics and their integration into recording media. Replacing traditional field-driven writing with all-optical switching (AOS) points to record-breaking data recording-rates with lower energy demands. A helicity-dependent AOS (HD-AOS) system could, in principle, write data on sub-picosecond timescales at nanometre lengths [2]. A key challenge is concentrating incident light into deep sub-wavelength volumes while maintaining strong circular polarisation. This necessitates an optical element that focuses light beyond the diffraction limit and preserves or induces helicity. This study addresses this by designing a plasmonic antenna that employs localised surface plasmon resonances (LSPRs) to tailor ultrafast laser light for AOS applications.

II. OPTICAL ANTENNA SIMULATION

We present a device concept simulated via the finite element method (FEM) in COMSOL Multiphysics: a nanoscale gold cross antenna with four-fold (C_{4v}) symmetry. Inspired by previous work on polarisation control with plasmonic nanoantennas [3], this design allows excitation with controlled polarisation and precise tuning of geometry, material permittivity, and wavelength. This reveals rich plasmonic behaviour, including strong local field enhancements—ideal for energy deposition into recording media—and high circularity, quantified via Stokes parameter analysis. Notably, the antenna shows quarter-wave plate behaviour, converting linearly polarised light to circularly polarised light and vice versa. This geometry is exhibited in Figure 1. Additionally, finite difference time domain (FDTD) methods have been used alongside COMSOL via FlexCompute's Tidy3D software to investigate the antenna's response to localised femtosecond laser pulses in the time domain rather than simple continuous waves in frequency domain analysis. In both cases, the antenna design is currently simulated atop an $L1_0$ -FePt slab layer where field-enhancement and polarisation control are monitored.

III. ANTENNA DESIGN

A. Cylindrical antenna

Initial antenna designs were influenced by the well known cross antenna shape proposed by Biagioni et al. in 2009 [3]. A general cross antenna device consists of four elongated gold nanorods set around a central feed gap where near-field enhancement is focused and maximised. Under illumination from near-infrared laser light, a longitudinal LSPR mode is induced along each rod, producing a strongly enhanced and confined light source at the centre of the antenna. Such enhancements are capable of maintaining the input polarisation state of light by way of the devices fourfold symmetry (C_{4v}) or indeed, tailoring the input polarisation to a desired output via symmetry-breaking ($C_{4v} \rightarrow C_{2v}$). While the cross-antenna design is promising in terms of polarisation control and strong enhancement at its centre, it remains a challenge to produce tailored near-field light (20 nm) beneath the device, where the recording media will sit and where optical switching is to occur. For this reason a secondary design, which is also present in literature [4], has been explored involving a far simpler gold cylinder device. It has been shown, in COMSOL, that a single cylinder with C_{4v} symmetry is also capable of conserving the polarisation state of light while also allowing for quarter wave plate behaviour if the cylinder is elongated into an elliptical geometry. While the maximum field enhancement observed is generally diminished compared to the cross-antenna, strong enhancement is observed beneath the cylinder antenna in the desired position.

B. Circularity

Circularity or the degree of circular polarisation is derived from the first and fourth Stokes parameters, S_0 and S_3 :

$$C = \frac{S_3}{S_0} = \frac{-2\Im\{\langle E_x \rangle \cdot \langle E_y^* \rangle\}}{\langle E_x \rangle^2 + \langle E_y \rangle^2}. \quad (1)$$

When the magnitudes of the electric field components are equal and have a phase difference of $\pm 90^\circ$, the conditions of circular polarization, C , become equal to ± 1 with the sign

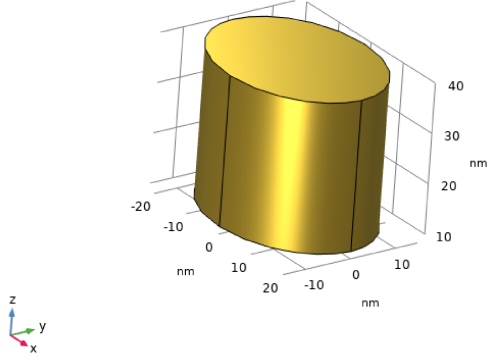


Figure 1. Elliptical cylinder antenna design.

indicating the handedness of the light. Using C as stated in 1, it becomes possible to evaluate the polarisation state of light at each point along a mesh grid. This convention can be utilised both in FEM and FDTD simulations, as shown in Figure 2.

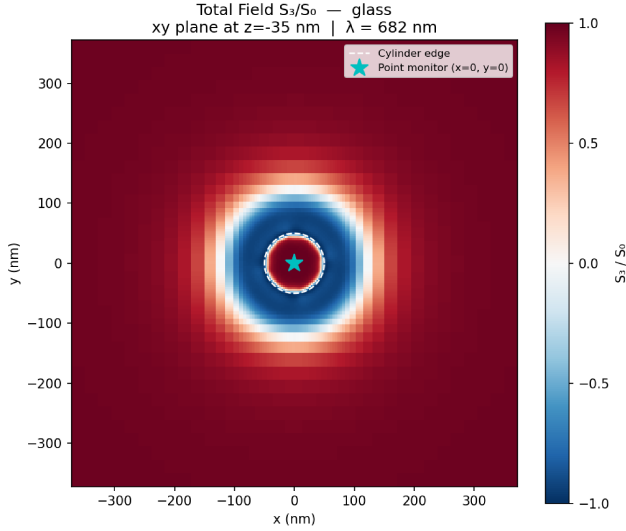


Figure 2. Strong conservation of circular polarisation beneath circular antenna upon illumination (Tidy3D simulation).

C. Thermal transport and electromagnetic heating

Gold's relatively low melting point and susceptibility to thermal expansion make it prone to morphological deformation (loss of C_4v) upon rapid Ohmic heating, which shifts the plasmonic resonance and degrades the helicity-selective near-field enhancement that drives switching. Unlike thermally assisted schemes such as HAMR, AOS is a non-thermal, angular-momentum-driven process; the thermal module therefore serves not as a description of the switching mechanism, but to monitor changes in gold's temperature-dependent permittivity and antenna geometry within safe fluence windows. Preliminary simulations have yielded antenna temperatures

consistent with comparable plasmonic devices such as the NFT in HAMR drives [5], and Figure 3 shows the resulting thermal profile during laser excitation.

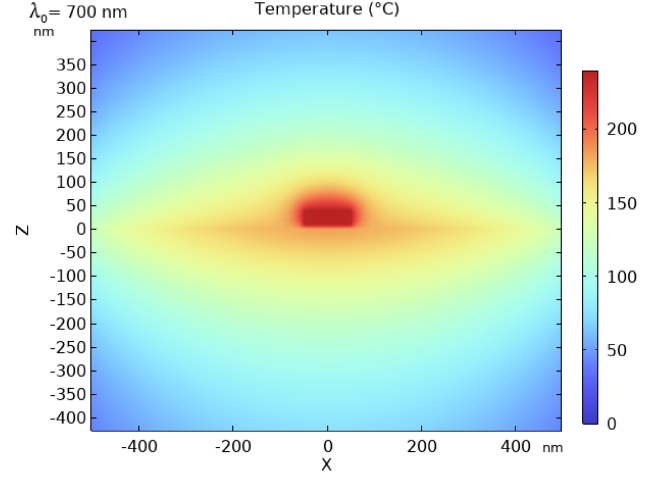


Figure 3. Thermal image of resonant antenna embedded in glass atop cobalt slab.

IV. LOOKING AHEAD

Future work will extend the present electromagnetic modelling framework to encompass fully coupled magnetic multilayer simulations. The existing COMSOL frequency-domain and Tidy3D FDTD models of the gold nanocylinder antenna will be expanded to incorporate the anisotropic magneto-optical permittivity tensor of $L1_0$ -FePt, enabling helicity-resolved near-field calculations within the recording layer and direct evaluation of the inverse Faraday effective field as a function of antenna geometry, polarisation state, and flying height. This coupled approach will subsequently be extended to include the full multilayer stack, assessing the influence of the dielectric overcoat and heat sink layers on both near-field enhancement and thermal confinement. Taken together, these simulations aim to establish predictive design rules for deterministic, sub-diffraction-limit, all-optical helicity-resolved magnetisation reversal in next-generation high-density magnetic recording.

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